(Simple) Linear Regression

Glantz
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- B. Variability about Regression Line
- C. Standard Error of Regression Coefficients

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Outline

- Linear regression
  - lm() function
- Correlation
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- Examples

Summary of Some Statistical Methods to Test Hypotheses

Scale of measurement	Type of experiment: Two treatment groups consisting of different individuals	Three or more treatment groups consisting of different individuals	Before and after a single treatment in the same individuals	Multiple treatments in the same individuals	Association between two variables
Interval (and drawn from normally distributed populations*)	Unpaired *t* test (Chapter 4)	Analysis of variance (Chapter 3)	Paired *t* test (Chapter 9)	Repeated-measures analysis of variance (Chapter 9)	Linear regression, Pearson product-moment correlation, or Bland-Altman analysis (Chapter 8)
Nominal	z Test Chi-square analysis-of-contingency table (Chapter 5)	Chi-square analysis-of-contingency table (Chapter 5)	McNemar's test (Chapter 9)	Cochrane Q†	Relative risk or odds ratio (Chapter 5) Logistic regression (slides)
Ordinal‡	Mann-Whitney rank-sum test (Chapter 10)	Kruskal-Wallis statistic (Chapter 10)	Wilcoxon signed-rank test (Chapter 10)	Friedman statistic (Chapter 10)	Spearman rank correlation (Chapter 8)
Survival time	Log-rank test or Gehan's test (Chapter 11)				

*If the assumption of normally distributed populations is not met, rank the observations and use the methods for data measured on an ordinal scale.
†Not covered in this text.
‡Or interval data that are not necessarily normally distributed.

I. Idea

200 Martians (**population**)

Height (cm) (X) ~ $N(40, 5^2)$

Weight (g) (Y) ~ $N(12, 2.5^2)$

<u>But Also:</u>

Mean weight of Martians at each height increases as height increases.

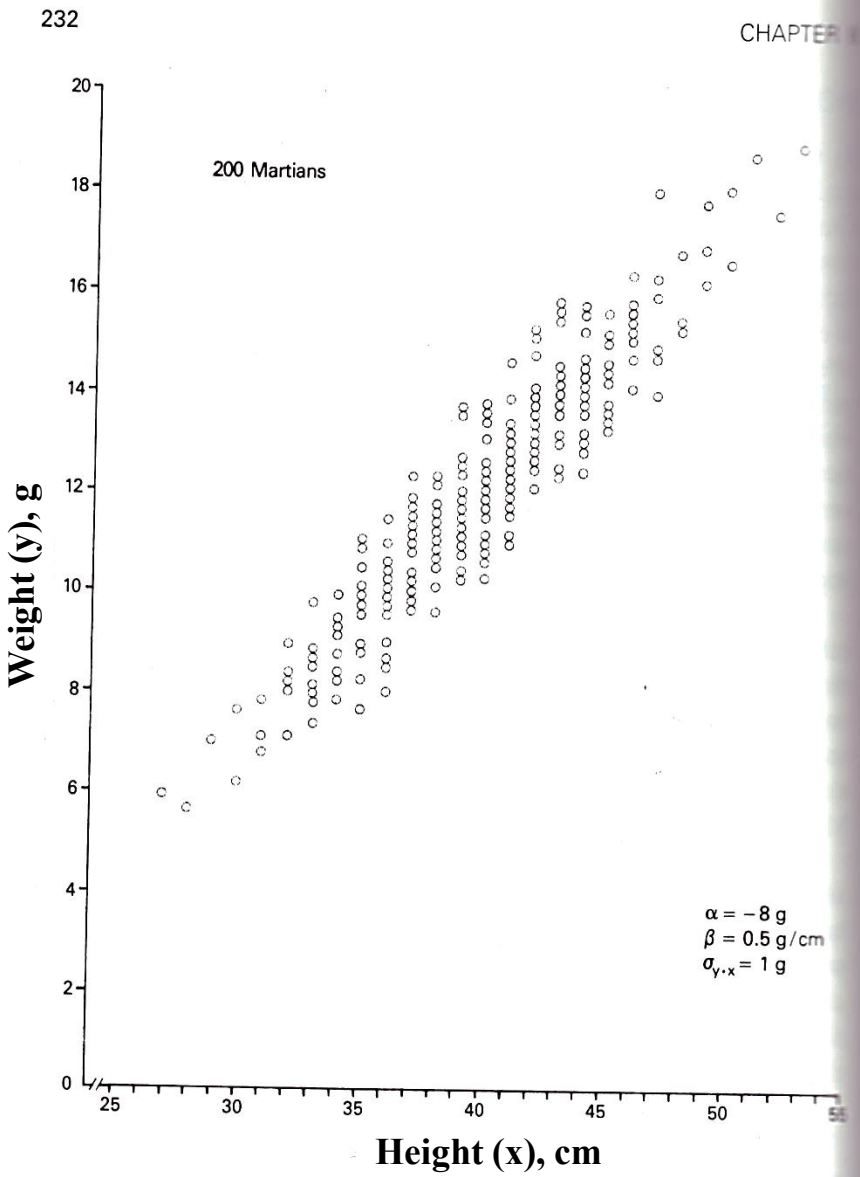
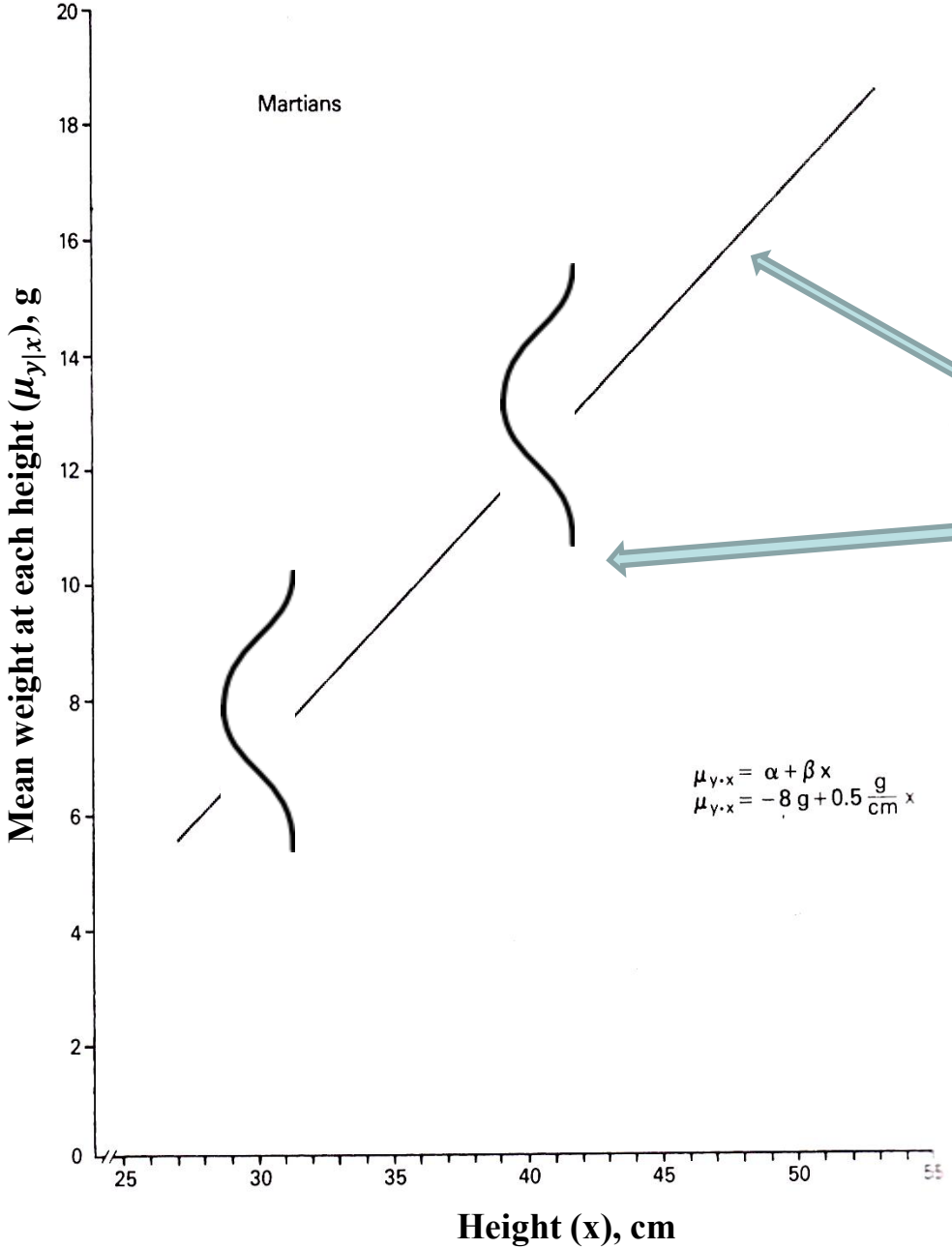


Figure 8-1 The relationship between height and weight in the population of 200 Martians, with each Martian represented by a circle. The weights at any given height follow a normal distribution. In addition, the mean weight of Martians at any given height increases linearly with height, and the variability in weight at any given height is the same regardless of height. A population must have these characteristics to be suitable for linear regression or correlation analysis.



The line of means for the population of Martians

X – independent variable

Y – dependent variable

$\mu_{y|x} = \alpha + \beta x$ line of means

$\sigma_{y|x}$ constant variability about line

given x $\quad y \sim N(\mu_{y|x}, \sigma^2_{y|x})$

Example

$$\mu_{y|x} = -8g + \left(0.5\ \mathrm{g/cm}\right)x$$

$$\sigma_{y|x} = 1g$$

Figure 8-2 The line of means for the population of Martians in Fig. 8-1.

Assumptions for the population

- The mean of the population of the dependent variable at a given value of the independent variable increases (or decreases) linearly as the independent variable increases

$$\mu_{y|x} = \alpha + \beta x$$

- For any given value of the independent variable, the possible values of the dependent variable are distributed **Normally**

$$y \sim N(\mu_{y|x}, \sigma^2_{y|x})$$

- The standard deviation of population of the dependent variable about its mean at any given value of the independent variable is the same for all values of the independent variable

$$\sigma_{y|x} = \text{constant}$$

II. Estimating Trends from a Sample

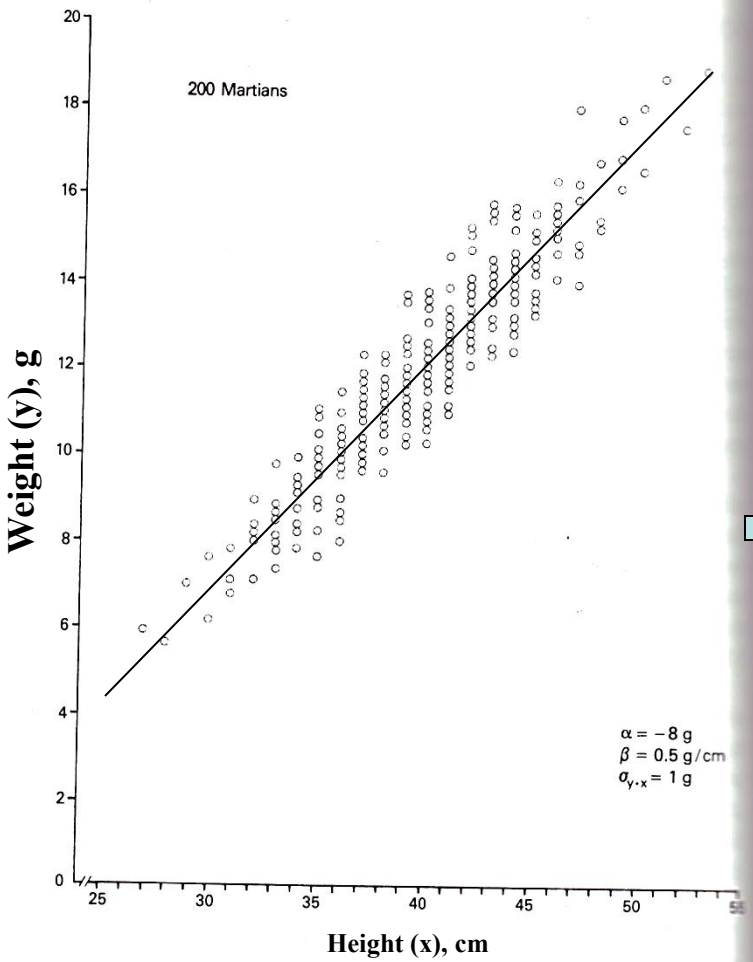


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A

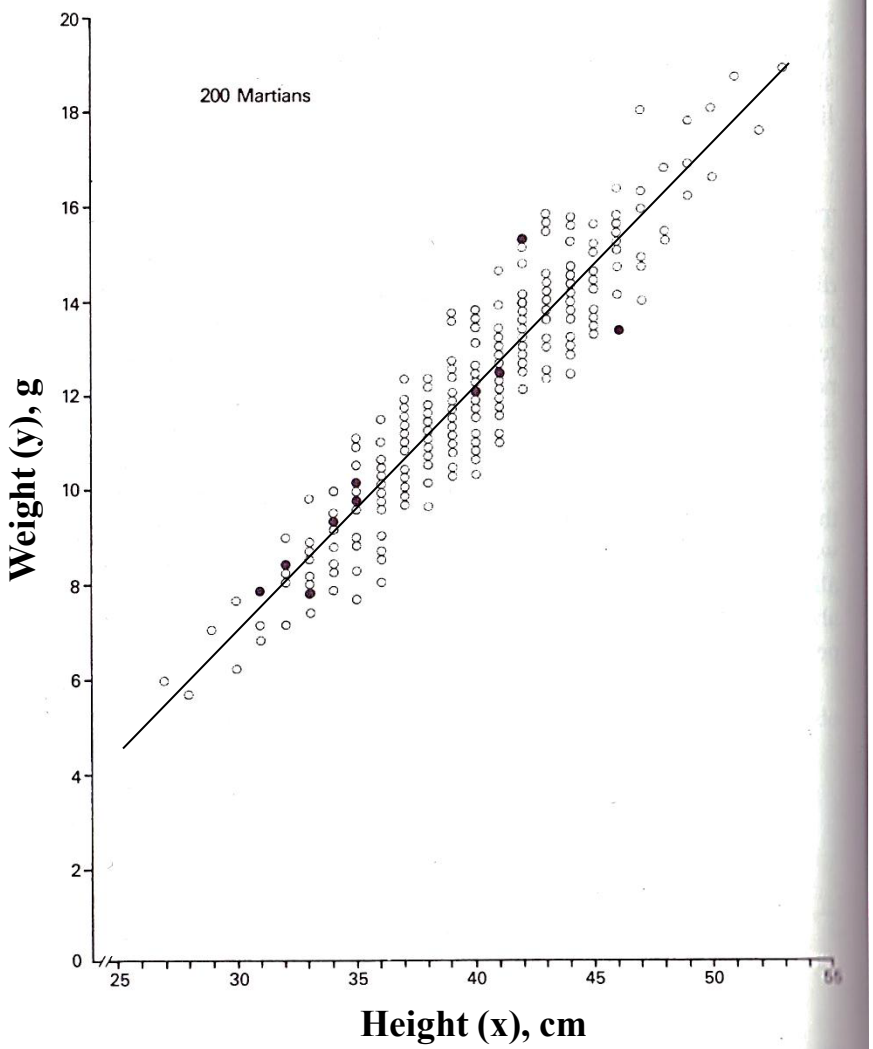


Figure 8-3 A random sample of 10 Martians, showing (*A*) the members of the population that were selected together with (*B*) the sample as it appears to the investigator.

Sample data from Marian Population

B

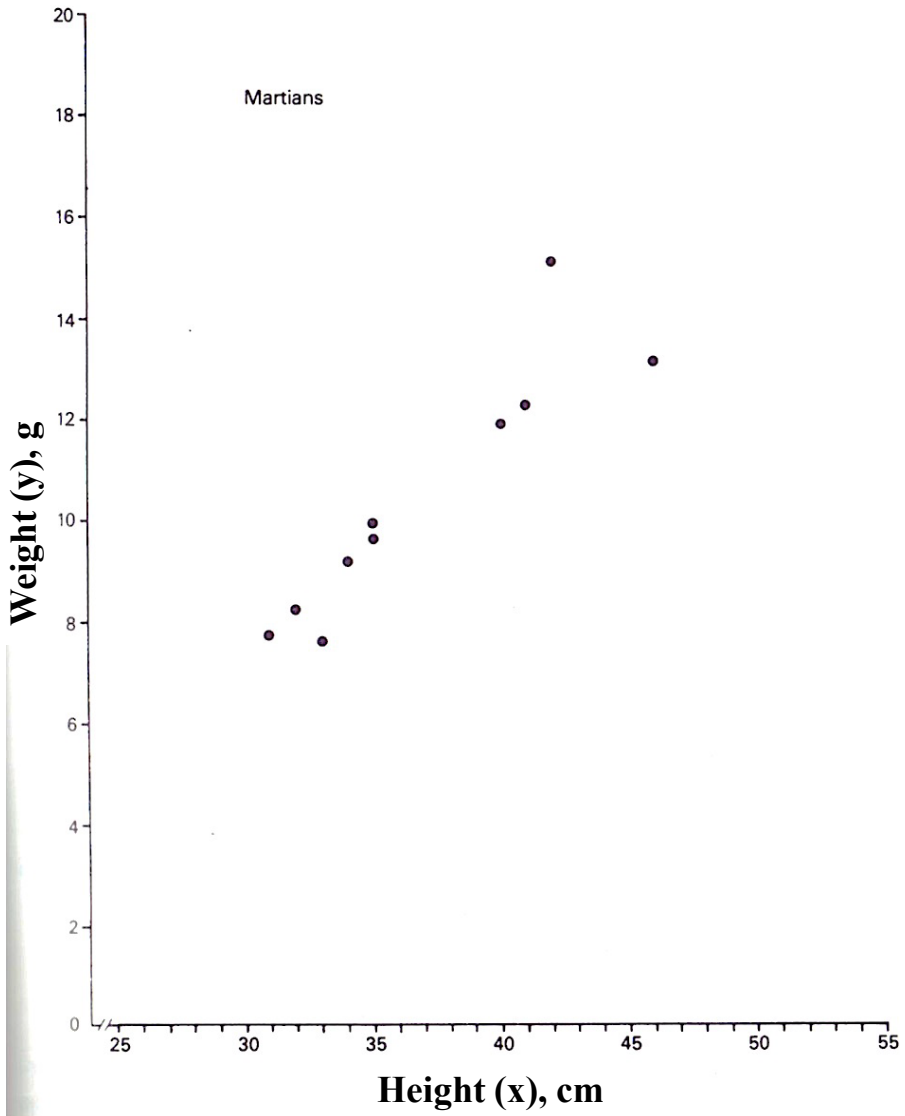


Figure 8-3 *(continued)*

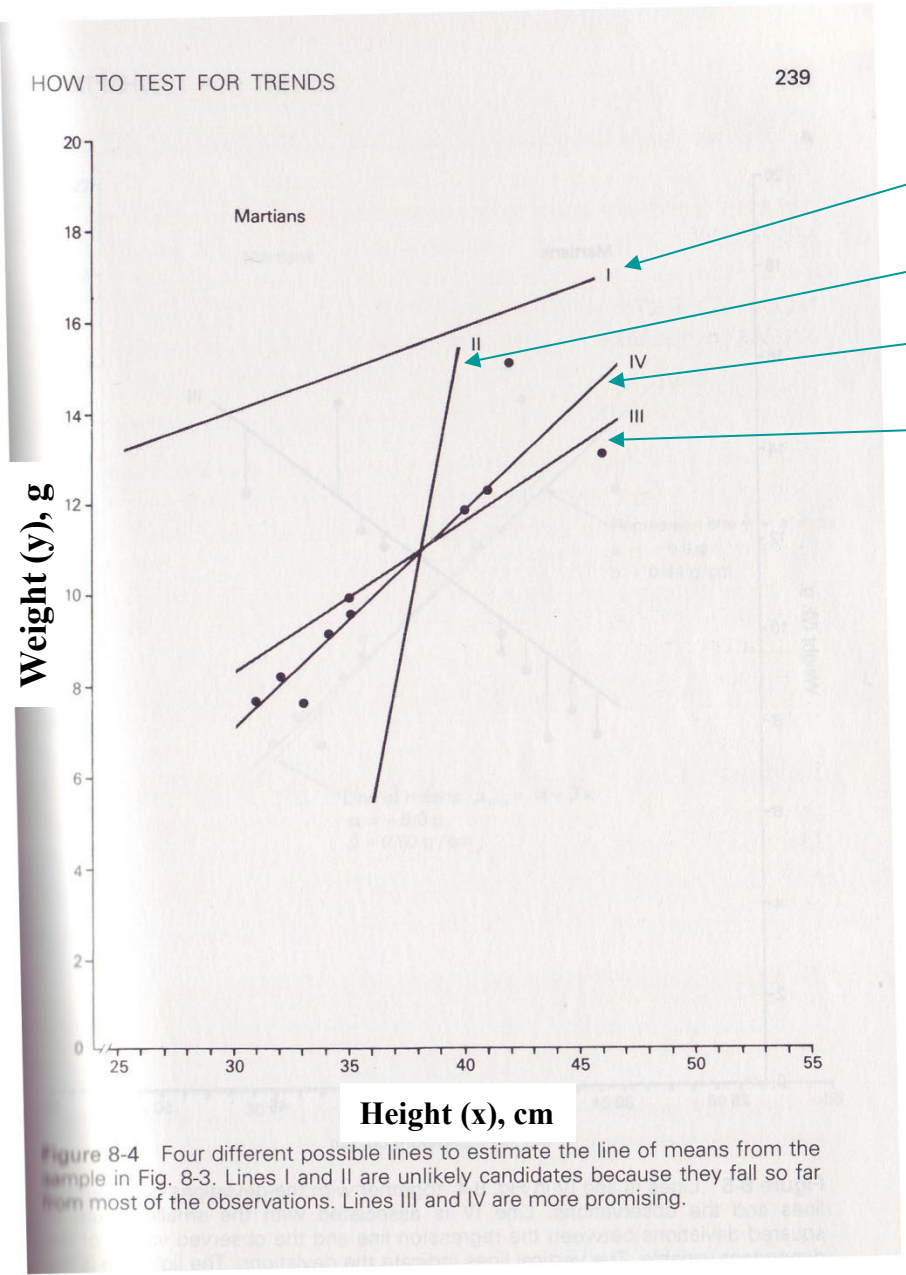


Figure 8-4 Four different possible lines to estimate the line of means from the sample in Fig. 8-3. Lines I and II are unlikely candidates because they fall so far from most of the observations. Lines III and IV are more promising.

I

II

IV

III

Which line is the best?

What does "Best" mean?

Best line means the line that minimizes the total variability between the data and the line.

Least Squares Estimation

- Gauss's Solution



The Schwarz portrait of Gauss (1803)

Gauss suggested:

"…the most probable value
of the unknown quantities
will be that in which the
sum of squares of the differences
between the actually observed and
computed values multiplied by
numbers that measure the
degree of precision
is a minimum."

A. Least-Squares Regression Line

A

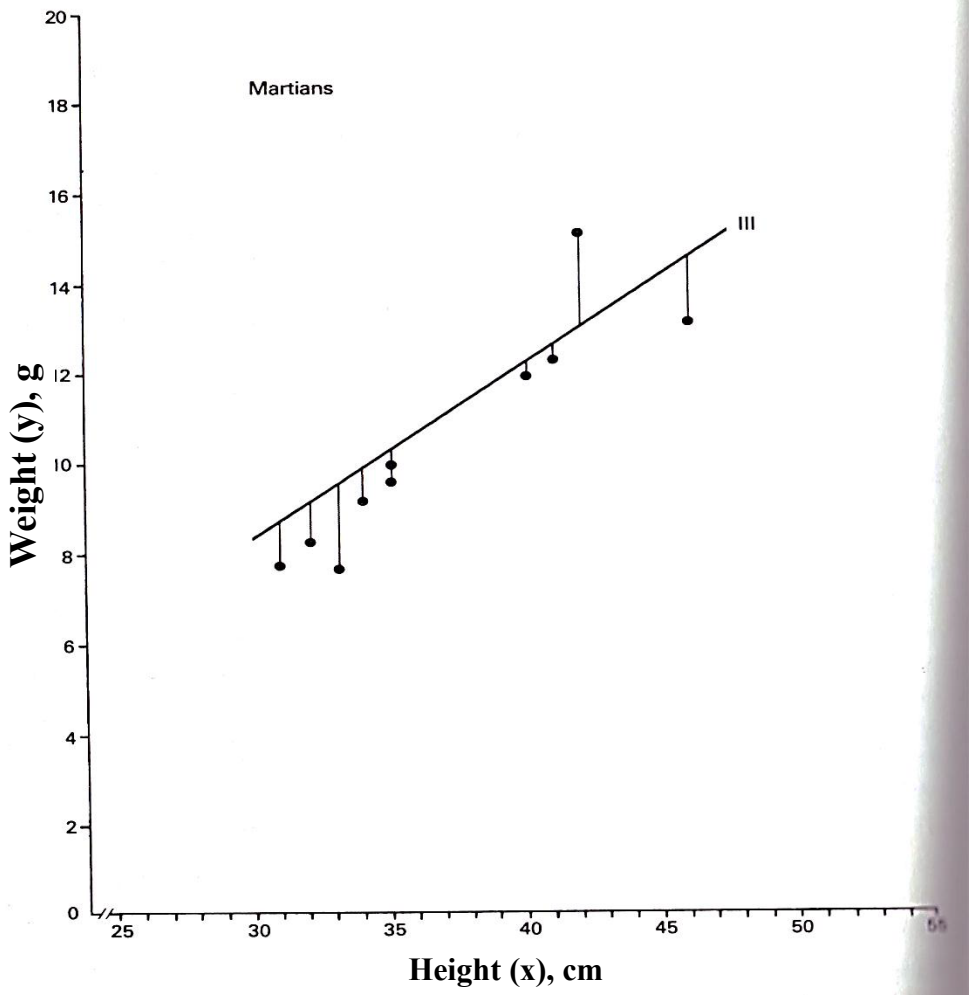


Figure 8-5 Lines III and IV in Fig. 8-4, together with the deviations between the lines and the observations. Line IV is associated with the smallest sum of squared deviations between the regression line and the observed values of the dependent variable. The vertical lines indicate the deviations. The light line is the line of means for the population of Martians in Fig. 8-1. The regression line approximates the line of means but does not precisely coincide with it. Line III in Fig. 8-4 is associated with larger deviations than line IV.

$$y_i = \alpha + \beta \cdot x_i + e_i$$

Observation — Independent Variable — Error

Minimize Sum of Squared Errors

$$SS = \sum_{i=1}^{n} \left(y_i - \hat{y}_i \right)^2$$

$$\hat{y}_i = a + b \cdot x_i$$

$$SS = \sum_{i=1}^{n} e_i^{\,2} = \sum_{i=1}^{n} \left(y_i - (a + bx_i) \right)^2$$

(x_i, y_i) $i = 1, \ldots, n$ Measurements of dependent and independent variables

Regression Line

$$\hat{y} = a + b \cdot x$$

Estimate of $\mu_{y|x}$

Estimate of α and β

$$\mu_{y|x} = \alpha + \beta x$$

For population

Sum of Squared Errors: $$SS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

$$SS = \sum_{i=1}^{n} (y_i - (a + b \cdot x_i))^2$$

Minimize SS w.r.t. a & b: $$\frac{\partial SS}{\partial a} = \frac{\partial SS}{\partial b} = 0$$

Solution $$SS = \sum_{i=1}^{n} (y_i - (a + b \cdot x_i))^2$$

$$\frac{\partial SS}{\partial a} = -2 \sum_{i=1}^{n} (y_i - (a + b \cdot x_i)) = 0$$

$$\frac{\partial SS}{\partial b} = -2 \sum_{i=1}^{n} (y_i - (a + b \cdot x_i)) \cdot x_i = 0$$

Arrange so a & b are on the same side

(1) $$\sum y_i = \sum_{i=1}^{n} (a + b \cdot x_i) = n \cdot a + b \cdot \sum x_i$$

(2) $$\sum y_i \cdot x_i = \sum_{i=1}^{n} (a \cdot x_i + b \cdot x_i^2) = a \cdot \sum x_i + b \cdot \sum x_i^2$$

From (1): $$a = \frac{1}{n} \sum y_i - b \cdot \frac{1}{n} \sum x_i \qquad (3)$$

Note also: $$a = \overline{y} - b \cdot \overline{x}$$

Substitute (3) into (2):

$$b = \frac{n\sum x_i y_i - \sum x_i \sum y_i}{n\sum x_i^2 - (\sum x_i)^2}$$

Substitute b into (1):

$$a = \frac{\sum y_i \sum x_i^2 - \sum x_i \sum x_i y_i}{n\sum x_i^2 - (\sum x_i)^2}$$

Can also use: $a = \overline{y} - b \cdot \overline{x}$

Table 8-1 Computation of Regression Line in Fig. 8-5*B*

Observed height *X*, cm	Observed weight *Y*, g	X^2, cm^2	XY, g·cm
31	7.8	961	241.8
32	8.3	1,024	265.6
33	7.6	1,089	250.8
34	9.1	1,156	309.4
35	9.6	1,225	336.0
35	9.8	1,225	343.0
40	11.8	1,600	472.0
41	12.1	1,681	496.1
42	14.7	1,764	617.4
46	13.0	2,116	598.0
369	103.8	13,841	3,930.1

*The calculations can be simplified by computing b first, then finding a from $a = \overline{Y} - b\overline{X}$, in which $\overline{X}$ and $\overline{Y}$ are the means of all observations of the independent and dependent variables, respectively.

$$n = 10$$

$$\sum x_i = 369 \text{ cm}$$

$$\sum y_i = 103.8 \text{ g}$$

$$\sum x_i^2 = 13{,}841 \text{ cm}^2$$

$$\sum x_i y_i = 3{,}930.1 \text{ g}\cdot\text{cm}$$

$$a = \frac{(103.8)(13841) - (369)(3930.1)}{10(13841) - (369)^2} = -6.0 \text{ g}$$

$$b = \frac{10(3930.1) - (369)(103.8)}{10(13841) - (369)^2} = 0.44 \text{ g/cm}$$

$$\hat{y} = -6.0 + 0.44x$$

B. Variability about Regression Line

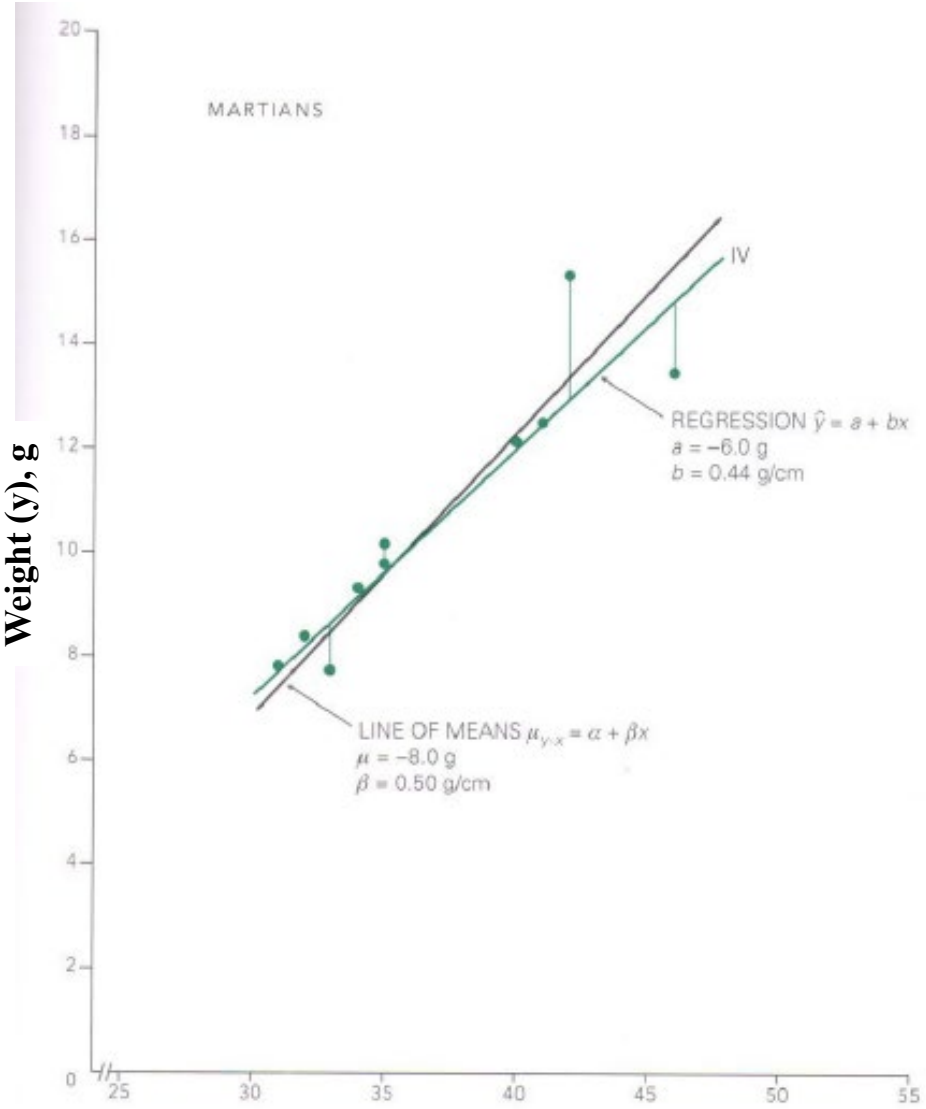


FIGURE 8-5. *(Continued)*

Regression line :

estimate of the line of means

$$\mu_{y|x} = \alpha + \beta x$$ **Actual/true line of means**

$$\mu_{y|x} = -8 + 0.5x$$

$$\sigma_{y|x} = 1 \text{ g}$$

$$\hat{y} = a + bx$$ **Regression line**

$$\hat{y} = -6.0 + 0.44x$$

Regression line: is an estimate of the line of means

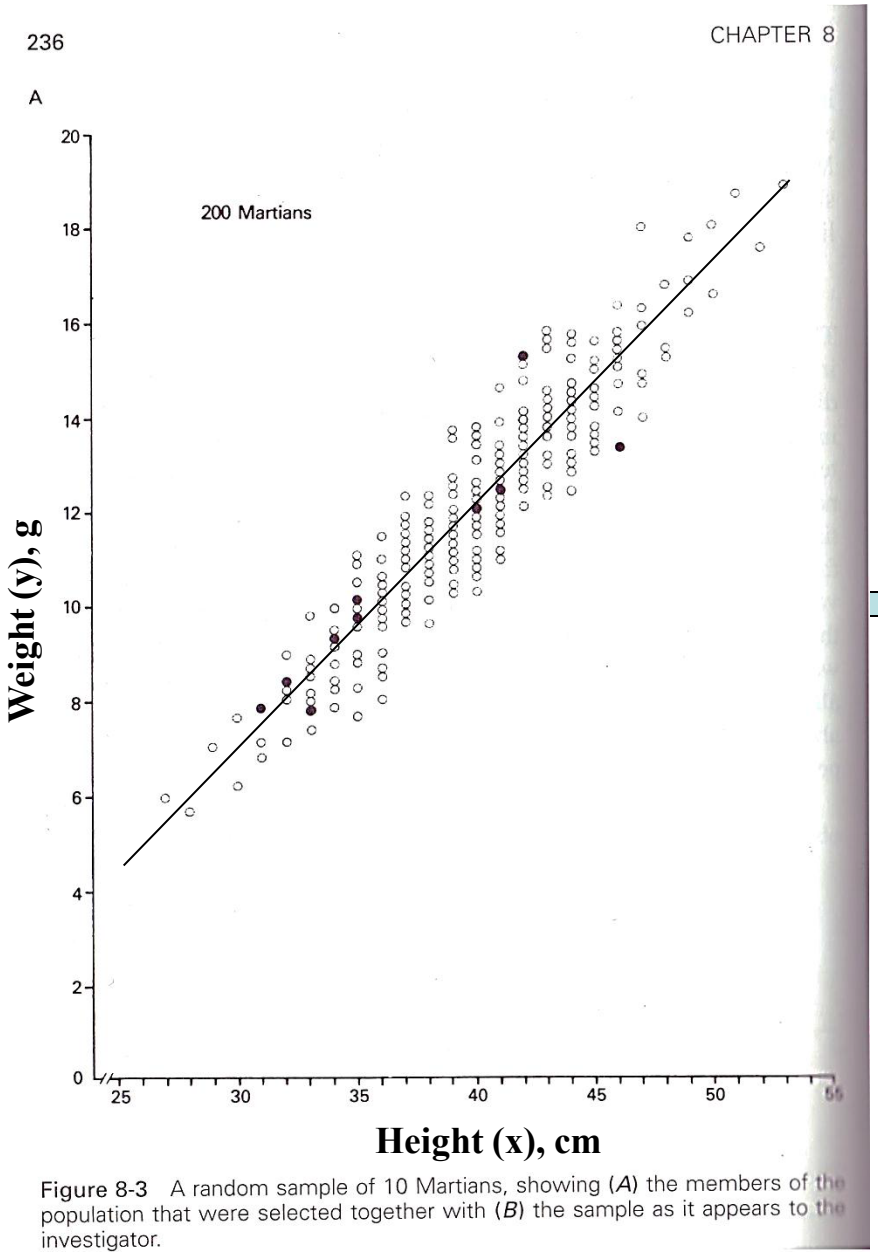


Figure 8-3 A random sample of 10 Martians, showing (*A*) the members of the population that were selected together with (*B*) the sample as it appears to the investigator.

Regression line: an estimate of the line of means

Also need to estimate the variability of population members about the line of the means, $\sigma_{y|x}$

$$s_{y|x}^2 = \frac{1}{n-2}\sum\left(y_i - \hat{y}_i\right)^2 = \frac{1}{n-2}\sum\left(y_i - (a + bx_i)\right)^2$$

Note: denominator is $(n-2)$, not n or $(n-1)$

You can show that the estimate of the variability around the line of means can also be written as

$$s_{y|x} = \sqrt{\frac{n-1}{n-2}\left(s_y^2 - b^2 s_x^2\right)}$$

$s_{y|x}$ "standard error of the estimate," also called "residual standard error"

Example (sample of 10 Martians)

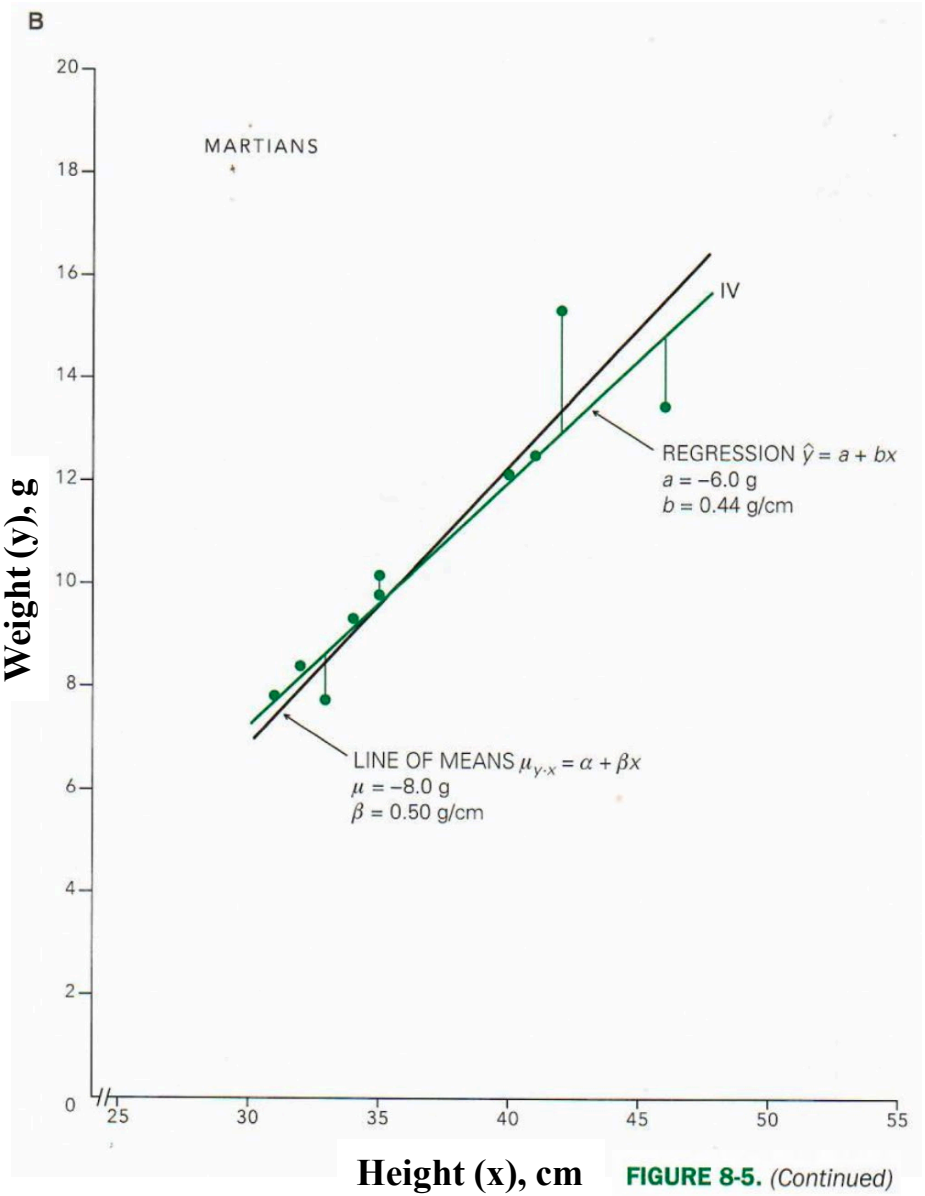


FIGURE 8-5. *(Continued)*

$$s_x = 5.0cm$$

$$n = 10$$

$$s_y = 2.4g$$

$$s_{y|x} = \sqrt{\frac{9}{8}[2.4^2 - 0.44^2(5.0^2)]} = 1.02g$$

$$\sigma_{y|x} = 1 \text{ g}$$

$s_{y|x}$ is the estimate of $\sigma_{y|x}$

C. Standard Error of Regression Coefficients

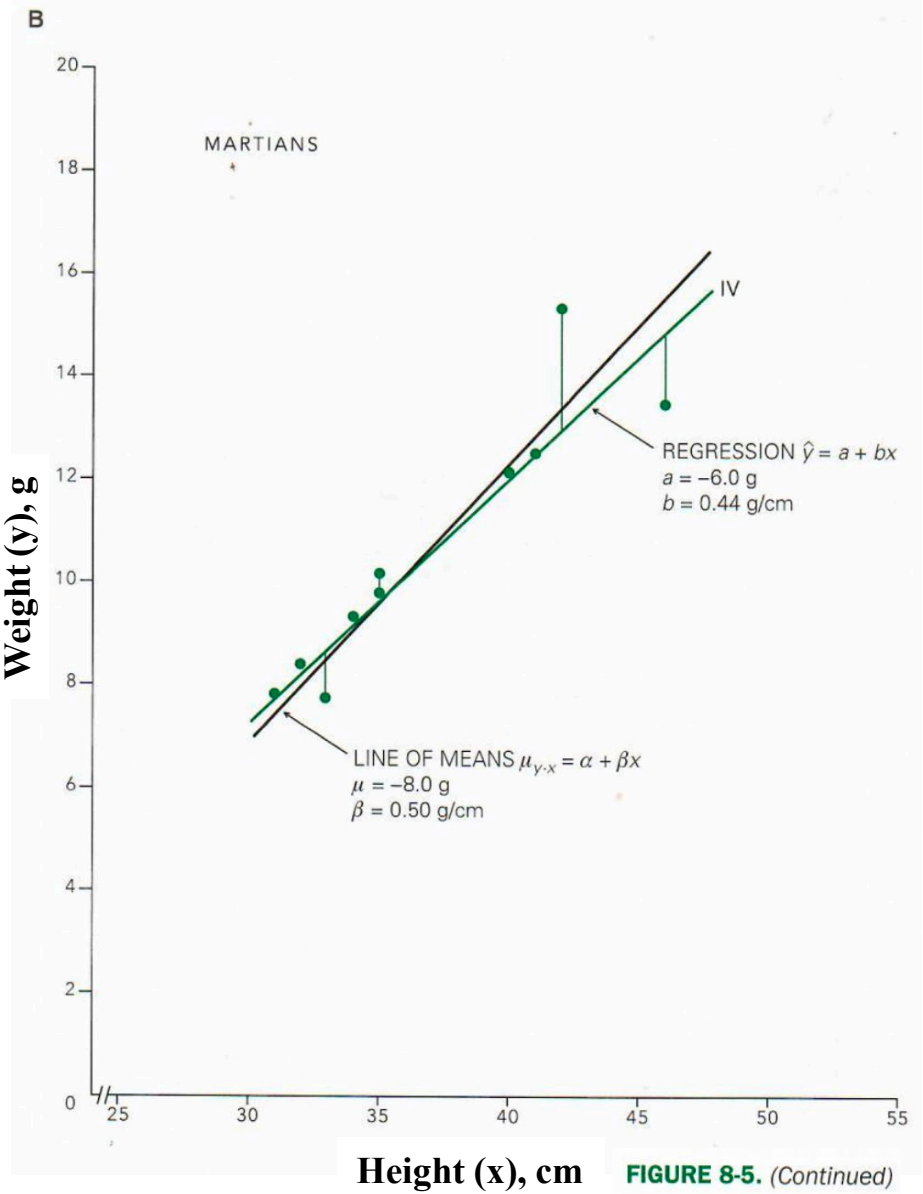


FIGURE 8-5. *(Continued)*

Standard error of a, the estimate of α

$$s_a = s_{y|x}\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{(n-1)s_x^2}}$$

Standard error of b, the estimate of β

$$s_b = \frac{1}{\sqrt{n-1}}\frac{s_{y|x}}{s_x}$$

In the example

$$s_a = (1.02g)\sqrt{\frac{1}{10} + \frac{(36.9)^2}{(10-1)(5.0)^2}} = 2.6g$$

$$s_b = \frac{1}{\sqrt{10-1}} \cdot \frac{1.02g}{5.0cm} = 0.064g/cm$$

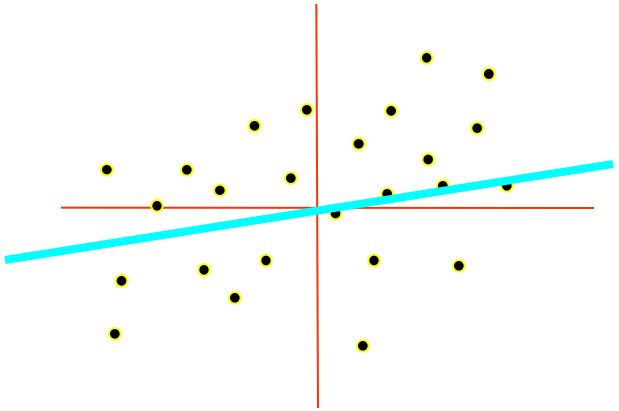
III. Hypothesis Testing:

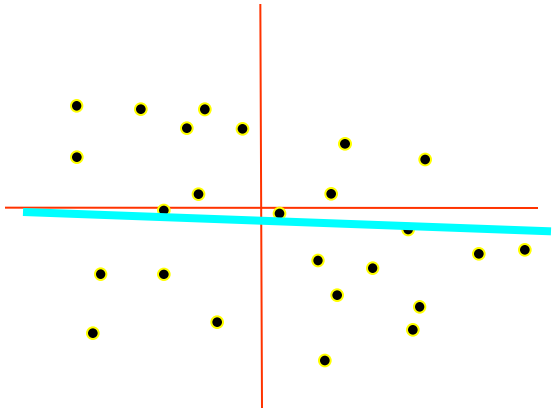
A. Significance of Trend (using t-statistic)

$$t = \frac{\textit{parameter estimate} - \textit{true value of poplulation parameter}}{\textit{standard error of parameter estimate}} = \frac{b - \beta}{s_b}$$

Is there a relationship between the variables???

$$H_0 : \beta = 0 \quad \Rightarrow \quad t = \frac{b}{s_b} \qquad v = n - 2$$





$$|t| > t_\alpha(v)?$$

(using two-tailed t table)

In the Example

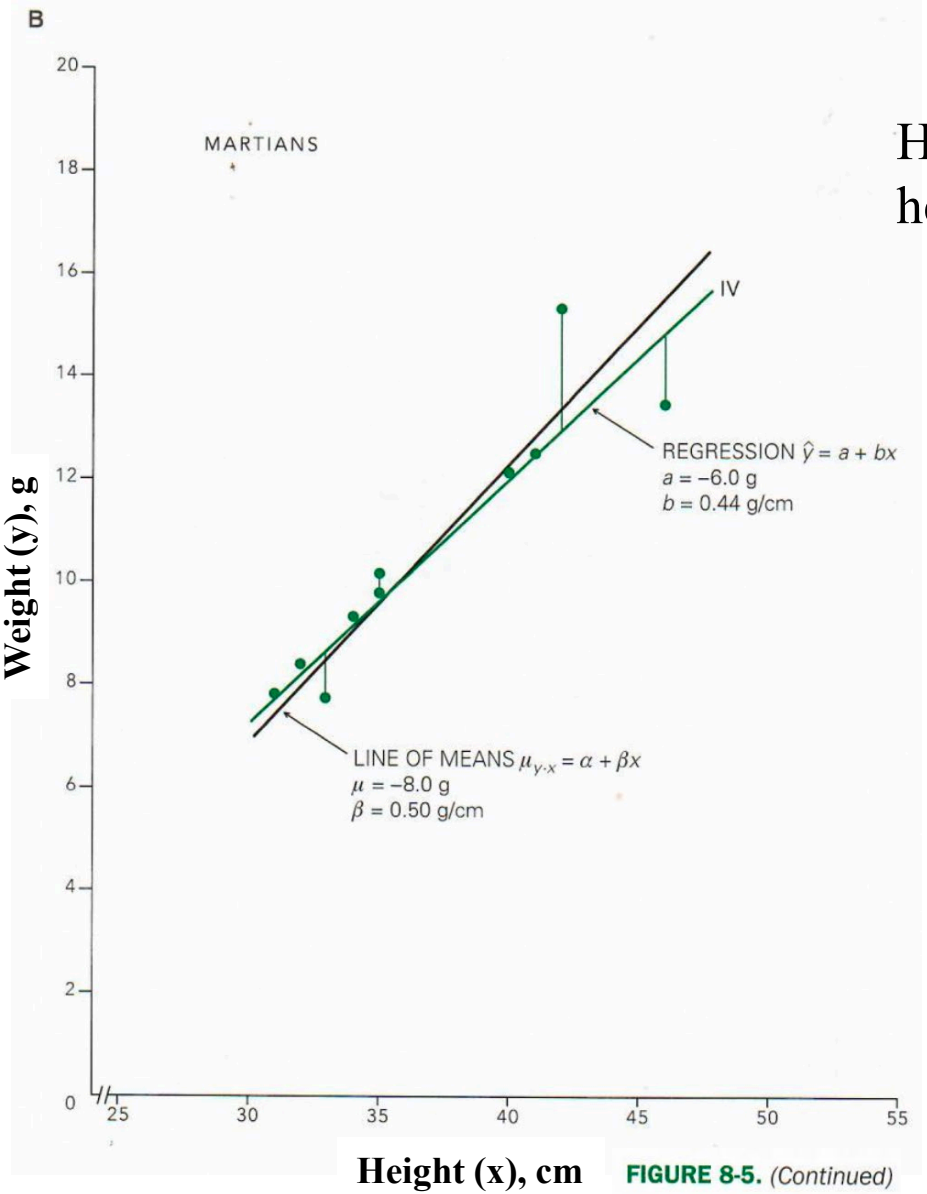


FIGURE 8-5. *(Continued)*

H_0 = There is no relationship between height and weight in Martians

$$b = 0.44\,g/cm \qquad s_b = 0.064\,g/cm$$

$$t = \frac{b}{s_b} = 6.875$$

$$v = n - 2 = 8 \qquad t_{0.001}(8) = 5.401$$

$$|t| > t_{0.001,crit}$$

Conclusion: Reject H_0 ($P < 0.001$)

B. CIs for β and α_{LOM}

$(1-\alpha)\cdot 100\%$ CI for β:

$$b - t_{\alpha}(\nu)\cdot s_b < \beta < b + t_{\alpha}(\nu)\cdot s_b$$

$(1-\alpha)\cdot 100\%$ CI for α_{LOM}:

$$a - t_{\alpha}(\nu)\cdot s_a < \alpha_{LOM} < a + t_{\alpha}(\nu)\cdot s_a$$

$$\nu = n - 2$$

C. CI for Line of Means (LOM)

$$\hat{y} = a + bx$$

Standard error for $\hat{y}$

$$s_{\hat{y}} = s_{y|x}\sqrt{\frac{1}{n} + \frac{(x-\bar{x})^2}{(n-1)s_x^2}}$$

Standard error for the LOM is a function of x

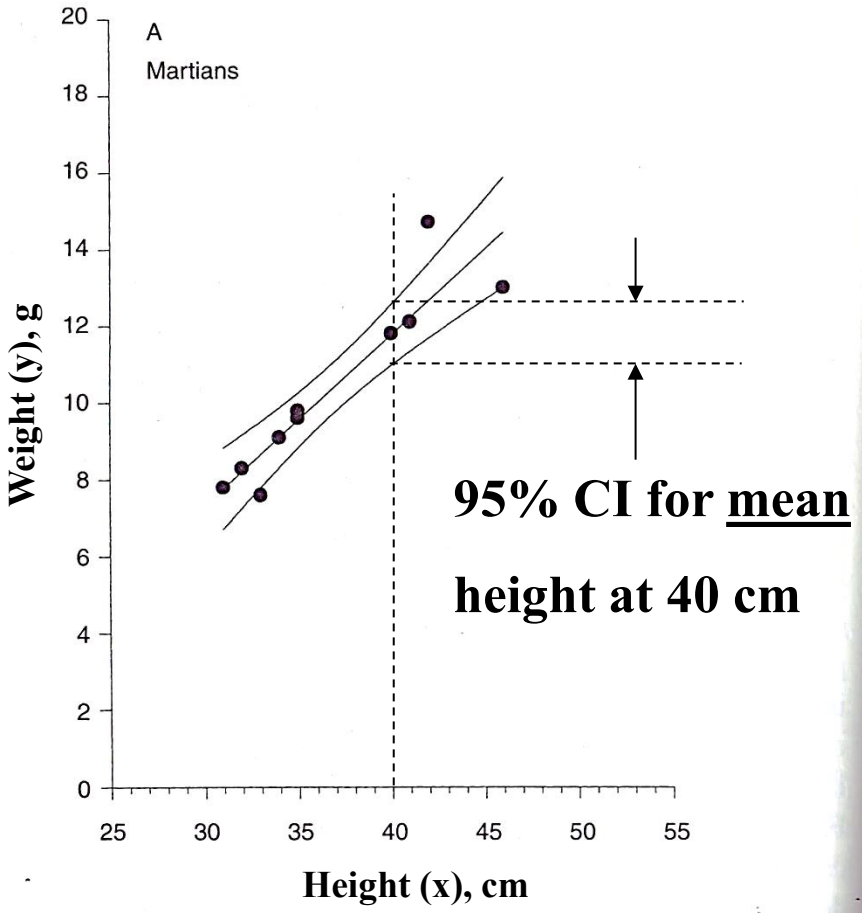


Figure 8-7 (*A*) The 95 percent confidence interval for the regression line relating Martian weight or height using the data in Fig. 8-3. (*B*) The 95 percent confidence interval for an additional observation of Martian weight at a given height. This is the confidence interval that should be used to estimate true weight from height to be 95 percent confident that the range includes the true weight.

CI for LOM

$$\hat{y} - t_\alpha(\nu)\cdot s_{\hat{y}} < \mu_{y|x} < \hat{y} + t_\alpha(\nu)\cdot s_{\hat{y}}$$

$$\nu = n - 2$$

D. CI for Single Observation

$$s_{Y_{new}} = \sqrt{s^2_{y|x} + s^2_{\hat{y}}}$$

$$s_{Y_{new}} = s_{y|x}\sqrt{1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{(n-1)s_x^2}}$$

(depends on x)

$$\hat{y} - t_\alpha(\nu)\cdot s_{Y_{new}} < y < \hat{y} + t_\alpha(\nu)\cdot s_{Y_{new}}$$

$$\nu = n - 2$$

Includes uncertainty due to variation in population and uncertainty in the estimate of the true LOM

95% CI for an Observation:

HOW TO TEST FOR TRENDS

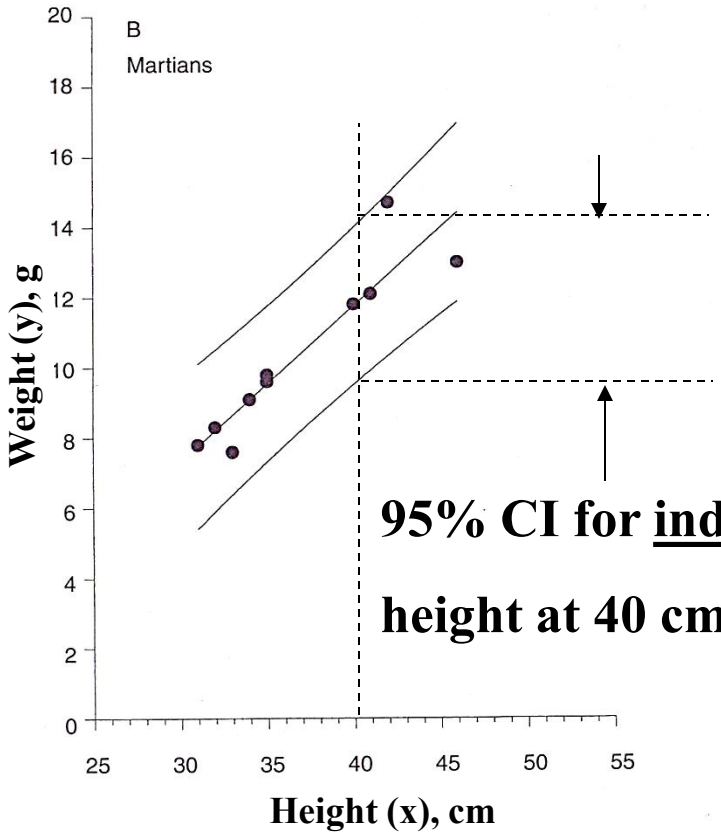


Figure 8-7 *(continued)*

IV. Comparing Two Regression Lines:

A. Hypothesis test of Slopes and Intercepts

Test for a difference in slope (using t-statistic)

$$t = \frac{difference\ of\ regression\ slopes}{standard\ error\ of\ difference\ of\ regression\ slopes} = \frac{b_1 - b_2}{s_{b_1 - b_2}}$$

$$\nu = n_1 + n_2 - 4$$

$$if\ \ |t| > t_{crit} = t_\alpha(\nu)\ \ reject\ H_0\ with\ P < \alpha$$

(using two-tailed t table)

If there are the **same number** of data points in each:

$$s_{b_1 - b_2} = \sqrt{{s_{b_1}}^2 + {s_{b_2}}^2} \qquad s_b = \frac{1}{\sqrt{n-1}} \frac{s_{y|x}}{s_x}$$

If there are a **different number** of points in each:

$$s_{b_1 - b_2} = \sqrt{\frac{s^2_{y|x_p}}{(n_1 - 1)s^2_{x_1}} + \frac{s^2_{y|x_p}}{(n_2 - 1)s^2_{x_2}}}$$

$$s^2_{y|x_p} = \frac{(n_1 - 2)s^2_{y|x_1} + (n_2 - 2)s^2_{y|x_2}}{n_1 + n_2 - 4}$$

Pooled estimate of variance about the two regression lines

Corresponding test for intercept

A. Example: Weight and Height in Males vs. Females Plutonians

Males

Height (inches)	Weight (lbs)
X	Y
62	198
64	207
71	228
73	233
68	217
65	212
64	204
64	201
71	228
67	218
66	210
65	212
68	219
69	226
68	222
73	237
70	222
67	219

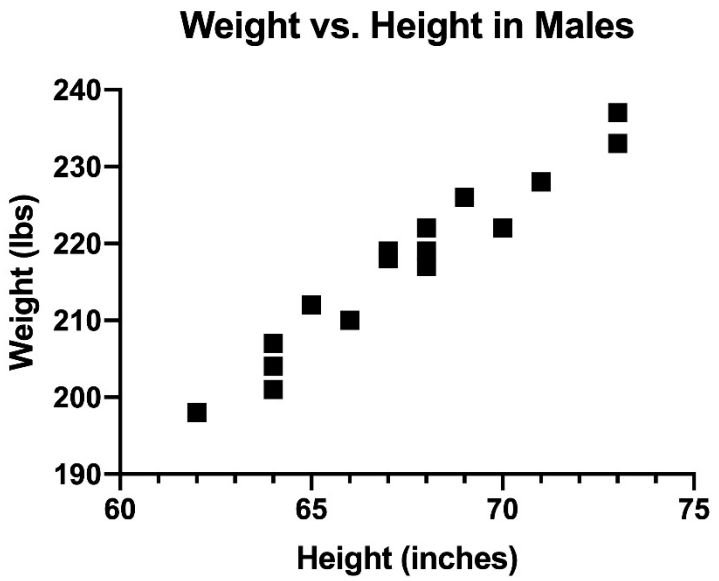
$n_{\mathrm{M}} = 18$

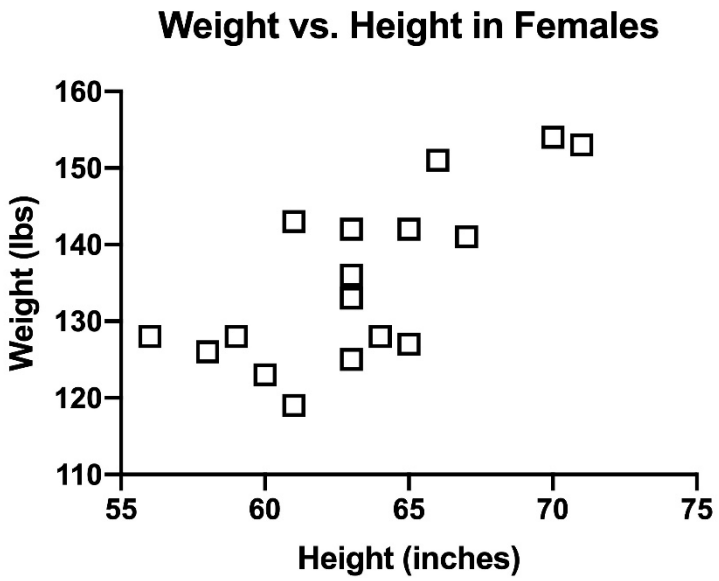
Females

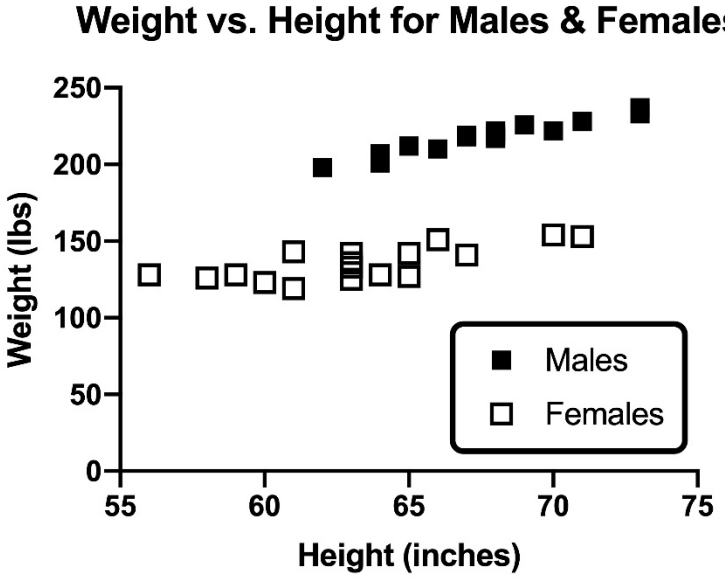
Height (inches)	Weight (lbs)
X	Y
61	119
56	128
64	128
61	143
63	125
71	153
63	142
58	126
63	136
59	128
66	151
67	141
63	133
70	154
65	127
65	142
63	133
60	123

$n_{\mathrm{F}} = 18$

Weight and Height in Males vs. Females: Exploratory Analysis of the Data







Weight and Height in Males vs. Females: Questions

1. Is there a significant relationship between weight and height in males?

2. Is there a significant relationship between weight and height in females?

3. If both are significant, is the relationship between weight and height for males different than that for females?
   A. Are slopes different for males and females?
   B. Are intercepts different for males and females?

How would you proceed to answer these questions?

1. Is there a significant relationship between weight and height in males?

Find the regression line between weight and height for <u>males</u>, then test if the slope of the line of means is different from 0.

```
11  # analysis for males
12  males <- subset(mydata, Sex =='M')
13  regression.m <- lm(formula = Weight.lbs ~ Height.in, data = males )
14  summary(regression.m)
```

Males

Height (inches)	Weight (lbs)
X	Y
62	198
64	207
71	228
73	233
68	217
65	212
64	204
64	201
71	228
67	218
66	210
65	212
68	219
69	226
68	222
73	237
70	222
67	219

$n_{\mathrm{M}} = 18$

1. Is there a significant relationship between weight and height in males?

Find the regression line between weight and height for <u>males</u>, then test if the slope of the line of means is different from 0.

```
> regression.m <- lm(formula = Weight.lbs ~ Height.in, data = males )
> summary(regression.m)

Call:
lm(formula = Weight.lbs ~ Height.in, data = males)

Residuals:
    Min      1Q  Median      3Q     Max
-4.8974 -1.9972 -0.4555  2.6777  3.6862

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  -4.2329    13.7901  -0.307    0.763
Height.in     3.2833     0.2041  16.088 2.66e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.711 on 16 degrees of freedom
Multiple R-squared:  0.9418,    Adjusted R-squared:  0.9381
F-statistic: 258.8 on 1 and 16 DF,  p-value: 2.661e-11
```

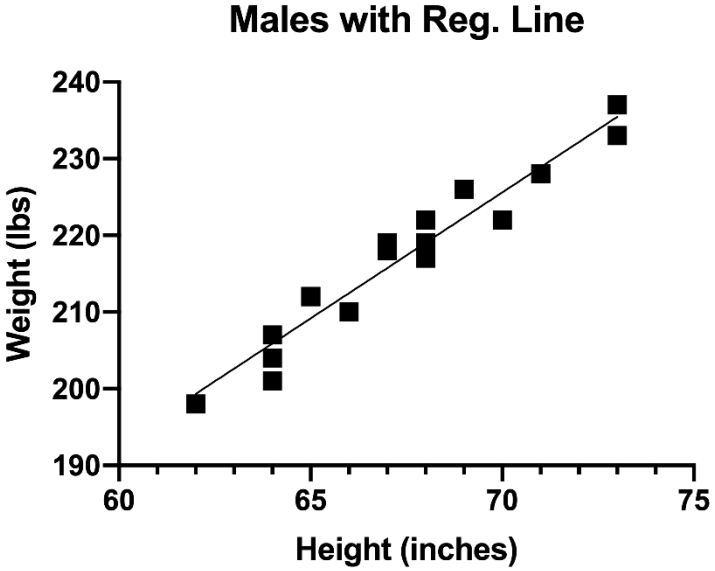
$\hat{y} = a + b \cdot x = -4.2329 + 3.2833 \cdot x$

$s_{y|x} = 2.711$

$H_0 : \beta = 0; \quad H_1 : \beta \neq 0;$

Reject H_0, $P \approx 0$

The regression line between weight and height for <u>males</u>



Since we concluded there is a significant relationship between weight and height for <u>males</u>, answer the following (on your own):

a. What are the standard errors of b_0 and b_1?
b. What are the 95% confidence intervals (CIs) for b_0 and b_1?
c. What is the correlation coefficient?
d. What is the mean weight of a male with a height of 68 inches?
e. What is the 95% CI for the mean weight in Part (d)?
f. What is the 95% CI for the weight of a male with a height of 68 inches?

2. Is there a significant relationship between weight and height in females?

Find the regression line between weight and height for <u>females</u>, then test if the slope of the line of means is different from 0.

```
38  # analysis for females
39  females <- subset(mydata, Sex =='F')
40  regression.f <- lm(formula = Weight.lbs ~ Height.in, data = females )
41  summary(regression.f)
```

Females

Height (inches)	Weight (lbs)
X	Y
61	119
56	128
64	128
61	143
63	125
71	153
63	142
58	126
63	136
59	128
66	151
67	141
63	133
70	154
65	127
65	142
63	133
60	123

$n_{\mathrm{F}} = 18$

2. Is there a significant relationship between weight and height in females?

Find the regression line between weight and height for <u>females</u>, then test if the slope of the line of means is different from 0.

```
> regression.f <- lm(formula = Weight.lbs ~ Height.in, data = females )
> summary(regression.f)

Call:
lm(formula = Weight.lbs ~ Height.in, data = females)

Residuals:
    Min      1Q  Median      3Q     Max
-11.718  -4.624   1.398   4.673  12.398

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   6.8349    29.8197   0.229  0.82161
Height.in     2.0290     0.4708   4.309  0.00054 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.491 on 16 degrees of freedom
Multiple R-squared:  0.5372,    Adjusted R-squared:  0.5082
F-statistic: 18.57 on 1 and 16 DF,  p-value: 0.0005401
```

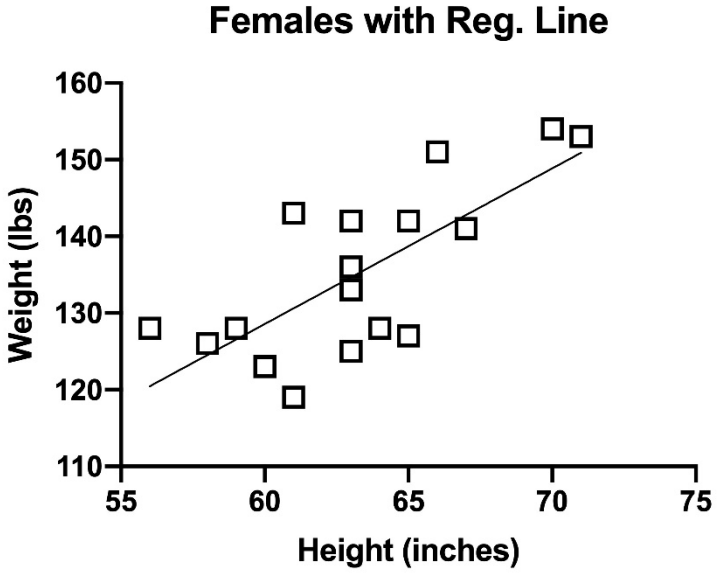
$\hat{y} = a + b \cdot x = 6.8349 + 2.0290 \cdot x$

$s_{y|x} = 7.491$

$H_0 : \beta = 0; \quad H_1 : \beta \neq 0;$

Reject H_0, $P < 0.001$

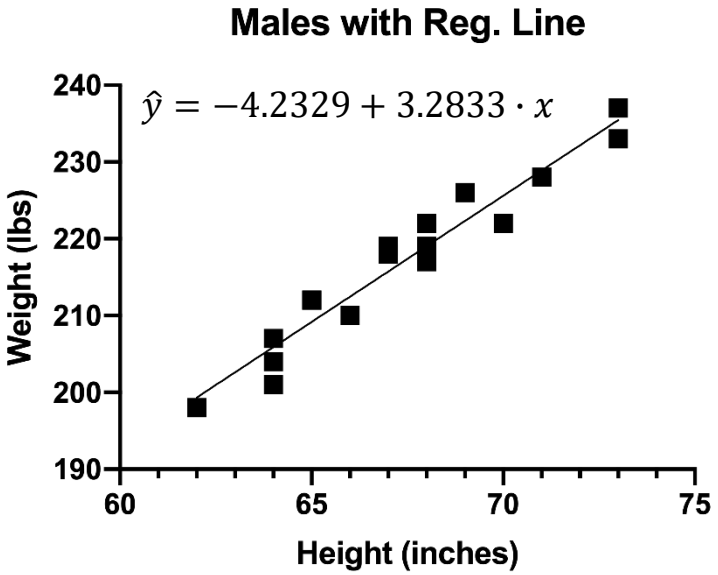
The regression line between weight and height for females

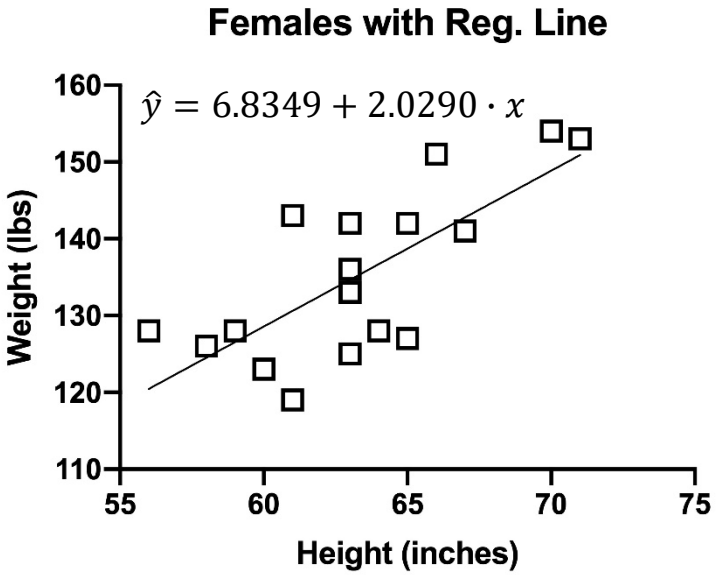


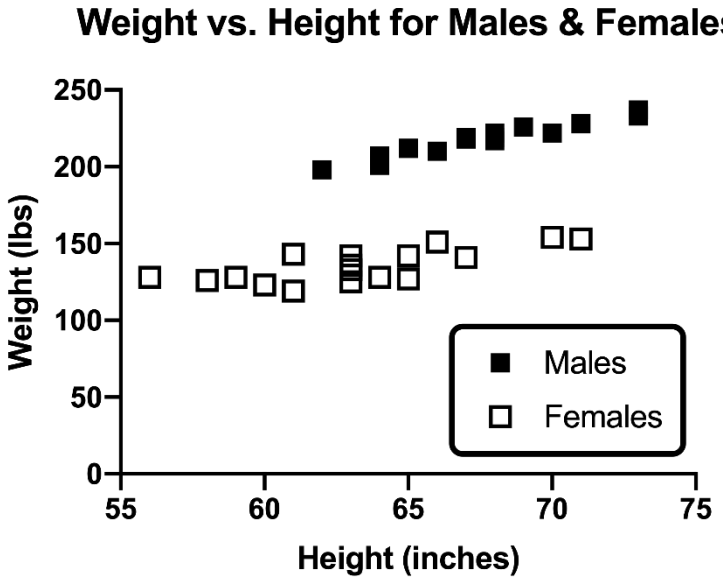
Since we concluded there is a significant relationship between weight and height for females, answer the following (on your own):

a. What are the standard errors of b_0 and b_1?
b. What are the 95% confidence intervals (CIs) for b_0 and b_1?
c. What is the correlation coefficient?
d. What is the mean weight of a female with a height of 68 inches?
e. What is the 95% CI for the mean weight in Part (d)?
f. What is the 95% CI for the weight of a female with a height of 68 inches?

3. Is the relationship between weight and height for <u>males</u> different than that for <u>females</u>?







IV. Correlation

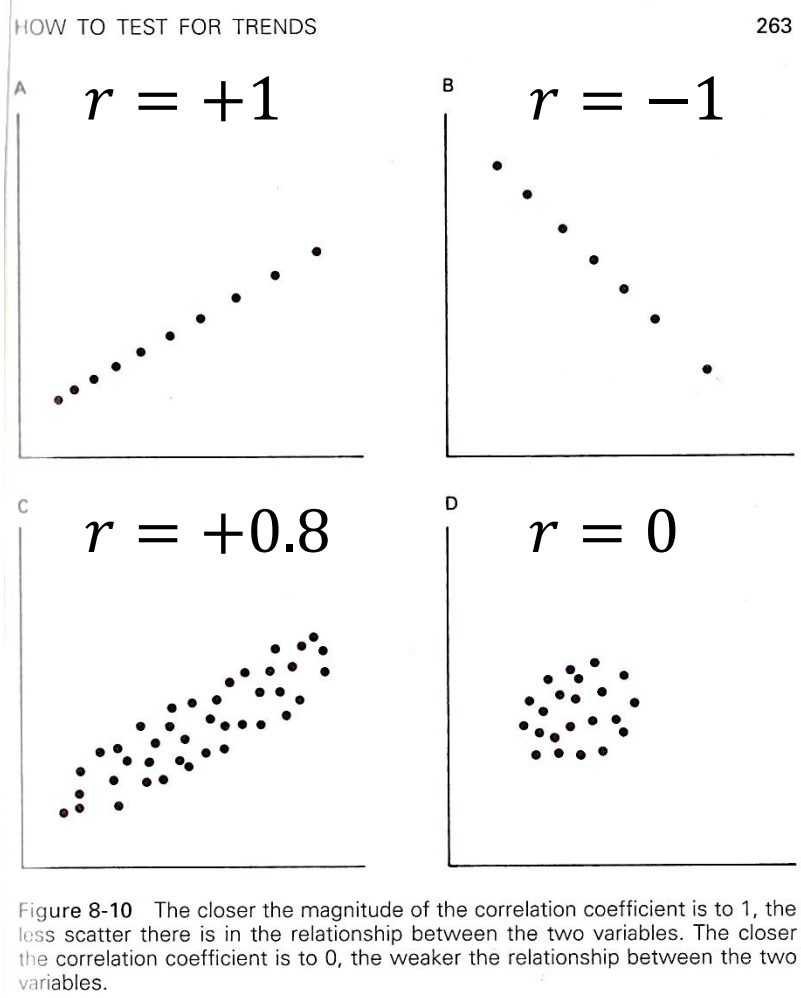


Figure 8-10 The closer the magnitude of the correlation coefficient is to 1, the less scatter there is in the relationship between the two variables. The closer the correlation coefficient is to 0, the weaker the relationship between the two variables.

The correlation coefficient (r) is a number between -1 and $+1$ and is used to quantify the strength of the relationship between two variables

The closer the magnitude is to 1 the stronger the relationship between the two variables

A value near zero indicates a weak correlation

Correlation versus Linear Regression

<u>Linear Regression</u>: Estimate of how, on average, a dependent variable changes as an independent variable changes. Also estimates the variability of the dependent variable about the LOM

<u>Correlation Coefficient</u>: Quantifies the strength of the association between two variables (does not require one variable to be dependent on the other)

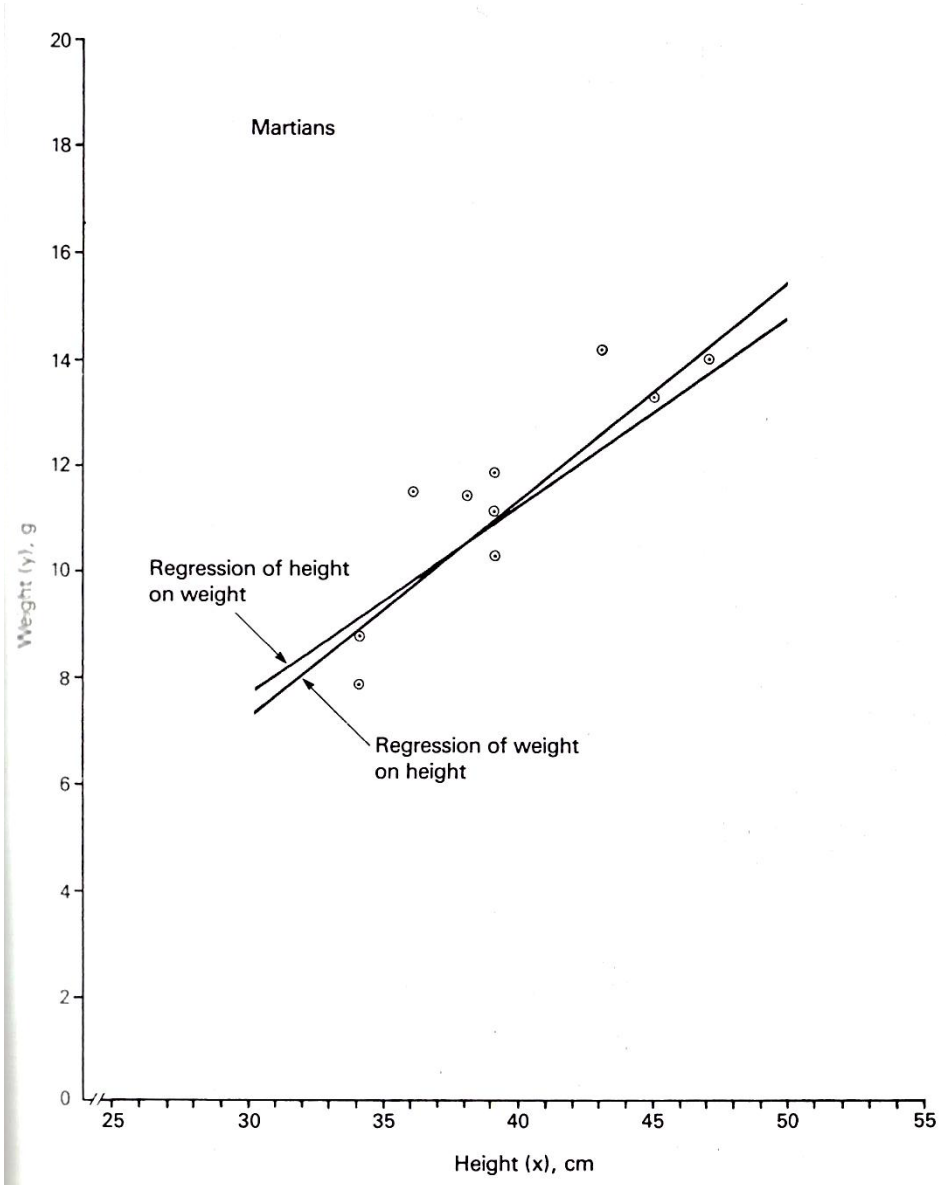


Figure 8-11 The regression of *y* on *x* yields a different regression line than the regression of *x* on *y* for the same data. The correlation coefficient is the same in either case.

A. Pearson Product-Moment Correlation Coefficient - r

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

$$r = \frac{\sqrt{\sum (x_i - \bar{x})^2}}{\sqrt{\sum (y_i - \bar{y})^2}} \cdot b$$

r close to $+1$	Variables increase together
r close to -1	Variables have an inverse relationship
r close to 0	Variables have no relationship

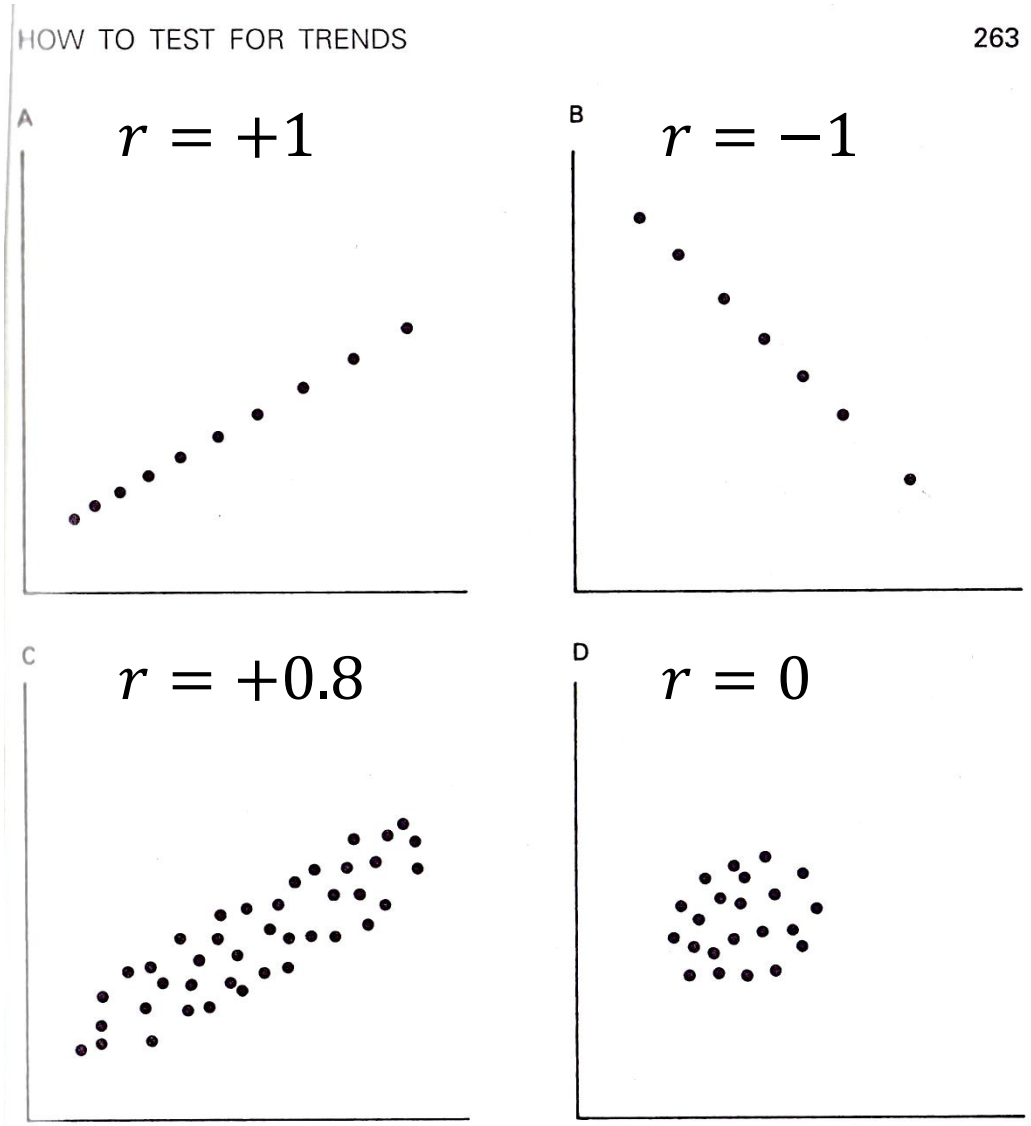


Figure 8-10 The closer the magnitude of the correlation coefficient is to 1, the less scatter there is in the relationship between the two variables. The closer the correlation coefficient is to 0, the weaker the relationship between the two variables.

Martian Example: Correlation Coefficient - *r*

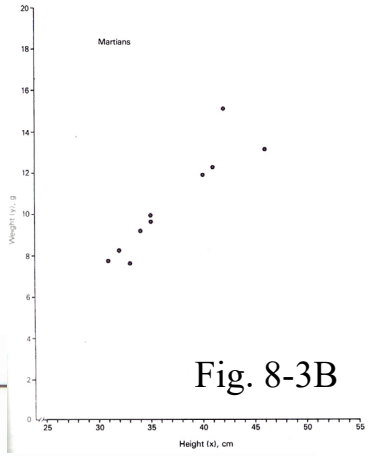


Fig. 8-3B

Table 8-3 Computation of Correlation Coefficient for Sample in Fig. 8-3*B*

Observed height X, cm	Observed weight Y, g	$(X-\overline{X})$, cm	$(Y-\overline{Y})$, g	$(X-\overline{X})(Y-\overline{Y})$, cm · g	$(X-\overline{X})^2$, cm^2	$(Y-\overline{Y})^2$, g^2
31	7.8	−5.9	−2.6	15.2	34.8	6.7
32	8.3	−4.9	−2.1	10.2	24.0	4.3
33	7.6	−3.9	−2.8	10.8	15.2	7.7
34	9.1	−2.9	−1.3	3.7	8.4	1.6
35	9.6	−1.9	−0.8	1.5	3.6	0.3
35	9.8	−1.9	−0.6	1.1	3.6	2.0
40	11.8	3.1	1.4	4.4	9.6	2.0
41	12.1	4.1	1.7	7.1	16.8	3.0
42	14.7	5.1	4.3	22.0	26.0	18.7
46	13.0	9.1	2.6	23.8	82.8	6.9
369	103.8	0.0	0.0	99.9	224.9	51.8

$$r = \frac{\sum(x-\bar{x})(y-\bar{y})}{\sqrt{\sum(x-\bar{x})^2 \sum(y-\bar{y})^2}} = \frac{99.9g \cdot cm}{\sqrt{(224.9cm^2)(51.8g^2)}} = 0.925$$

V. Correlation

B. Hypothesis Test of Correlation Coefficient

H_0: there is no relationship between the two variables (i.e., $r = 0$)

Test for significance of r (using t-statistic)

$$t = r\sqrt{\frac{n-2}{1-r^2}} \qquad \nu = n-2$$

Compare t-statistic to critical t value:

$$if \quad |t| > t_{crit} = t_{\alpha}(\nu) \quad reject\ H_0\ with\ P < \alpha$$

(using two-tailed t table)

VI. `lm()` Function in R

Linear Regression *lm()* function

How to use the *lm()* and associated R functions to perform simple linear regression.

```
results <- lm(formula = Y ~ X, data = dataname)
print(results)
summary(results)
coefficients(results, ...) # also coef()
confint(results, level = 0.95, ...)
residuals(results, ...)
plot() # e.g., look at data vs. regression line
predict(results, ...)
cor(x = dataname$X, y = dataname$Y, ...)
cor.test(x = dataname$X, y = dataname$Y, ...)
```

"lm": linear model

VI. `lm()` Function in R

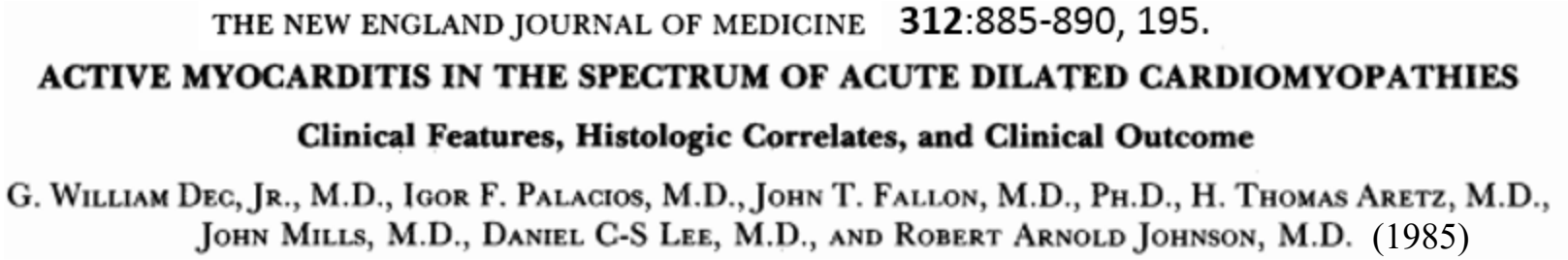
Linear Regression *lm()* function examples

Example #1: Martian height versus weight. Use R to reproduce the results presented in the lecture that were done "by hand."

1. What are the values of the slope and intercept of the regression line? Compare them to the values calculated in Lecture Slide 14.
2. What is the residual standard error (standard error of the estimate)? Note that this value is different from that on Lecture Slide 18 because we rounded while doing the calculations in the slides (i.e., we introduced small rounding errors that compounded).
3. What are the standard errors of the estimates for the slope and intercept? See Lecture Slide 19.
4. Is there a significant relationship between the Martian heights and weights? See Lecture Slide 21.
5. Construct a 95% confidence interval for the line of means for Martians who are 40 cm tall. See Lecture Slide 23.
6. Construct a 95% confidence interval for the weight of Martians who are 40 cm tall. See Lecture Slide 24.

Linear Regression *lm()* function examples

Example # 2: In patients diagnosed with acute dilated cardiomyopathy resulting from myocarditis, examine the relationship between left ventricular ejection fraction (LVEF) and age.



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ACTIVE MYOCARDITIS IN THE SPECTRUM OF ACUTE DILATED CARDIOMYOPATHIES

Clinical Features, Histologic Correlates, and Clinical Outcome

G. WILLIAM DEC, JR., M.D., IGOR F. PALACIOS, M.D., JOHN T. FALLON, M.D., PH.D., H. THOMAS ARETZ, M.D., JOHN MILLS, M.D., DANIEL C-S LEE, M.D., AND ROBERT ARNOLD JOHNSON, M.D. (1985)

1. What are the values of the slope and intercept of the regression line of LVEF on age?
2. Is there a significant relationship between LVEF and age in this population?
3. What is the average LVEF of a 45-year-old patient?
4. What are the standard errors of the slope and intercept?
5. Provide a 95% CI of the slope and intercept.
6. What is the correlation coefficient between LVEF and age?
7. Is the correlation coefficient different from 0?